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For chromium - 24

Expected configuration :

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^4\ 4s^2$

Actual configuration :

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$

For copper - 29

Expected configuration :

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^9\ 4s^2$

Actual configuration :

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^1$

The reason for above observed configuration is that fully filled orbitals and half filled orbitals have been found to have extra stability. In other words, p^3, p^6, d^5, d^{10}, f^7 and f^{14} configurations are more stable than p^2, p^5, d^4, d^9, f^6 and f^{13}. Due to this stability, one of the 4s electrons occupies the 3d orbital in chromium and copper to attain the half filled and the completely filled configurations respectively.

Evaluate Yourself

7. Explain the meaning of the symbol $4f^2$. Write all the four quantum numbers for these electrons.

2.6.5 Stability of half filled and completely filled orbitals:

The exactly half filled and completely filled orbitals have greater stability than other partially filled configurations in degenerate orbitals. This can be explained on the basis of symmetry and exchange energy. For example chromium has the electronic configuration of $[Ar]3d^5\ 4s^1$ and not $[Ar]3d^4\ 4s^2$ due to the symmetrical distribution and exchange energies of d electrons.

Symmetrical distribution of electron:

Symmetry leads to stability. The half filled and fully filled configurations have symmetrical distribution of electrons (Figure 2.13) and hence they are more stable than the unsymmetrical configurations.

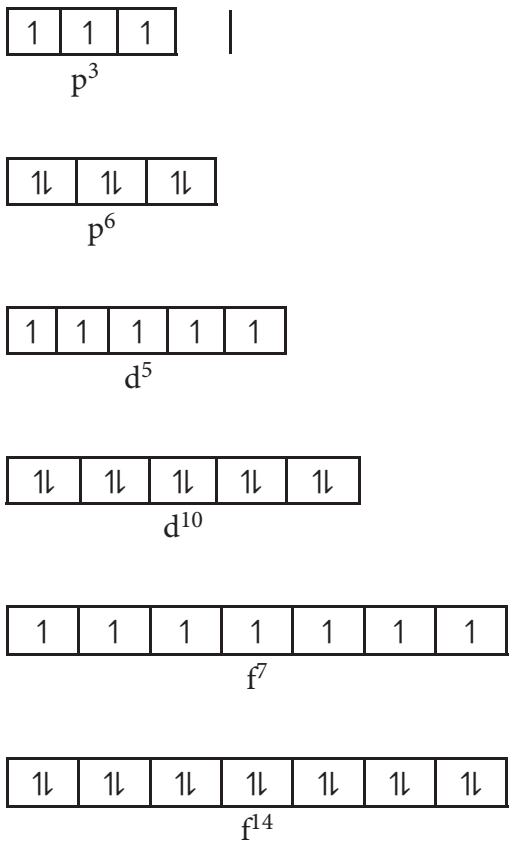


Figure 2.13 Half filled and fully filled p, d and f orbitals

The degenerate orbitals such as p_x, p_y and p_z have equal energies and their orientation in space are different as shown